

Exploring Practices, Challenges, and Design Implications for Citation Foraging, Management, and Synthesis

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Abstract

Citations play a crucial role in reinforcing knowledge construction, sparking new ideas, and fostering science communication. However, researchers often encounter difficulties in foraging, managing, and synthesizing citations amid the rapid growth of academic publications. To better understand researchers' practices and challenges in citation related activities, we conducted semi-structured interviews with 12 researchers in the fields of HCI and AI. Our key findings include: (1) researchers are unable to fully track and digest all relevant work; (2) current citation management tools fall short of meeting researchers' needs; (3) "cherry-picking" (write the statement first, then search for supporting references to strengthen its credibility and accuracy) is a common practice in citation synthesis; and (4) citation foraging, management, and synthesis workflows are disconnected and lack consistency. Our design implications provide insight into the development of interactive systems that more effectively support researchers in their citation activities.

CCS Concepts

• Human-centered computing → Empirical studies in HCI.

Keywords

Citation Foraging, Citation Management, Citation Synthesis, Bibliography

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1 Introduction

Citation is the foundation of academic work; researchers draw on prior knowledge from citations, conducting literature reviews to filter relevant papers that help construct the knowledge space and provide inspiration [5, 16]; Meanwhile, researchers use citations as justifications in their own papers to support their statements for the introduction, comparison, use, and extension of related work [3, 50]. Consequently, researchers invest significant time and effort in *foraging* latest publications [31], *managing* their bibliographies [5], and *synthesizing* citations to refer to them in their own work [3].

These problems are amplified by the ever-increasing volume of research papers being published nowadays [17, 18, 28], especially in dynamic research fields. For instance, NeurIPS (formerly NIPS), a top-tier international AI conference, has seen the number of accepted papers increase tenfold over 10 years, from 403 papers in 2015 to 4,037 in 2024 [14]). In a similar way, the number of papers published in the ACM CHI proceedings more than doubled over 10 years, increasing from 484 in 2015 to 1,060 in 2024 [8, 51]. This further complicates the ability of researchers to stay updated with the latest research and to thoughtfully select citations from an ever-expanding body of research [17]. This issue is exacerbated by the trend of preprints, which increases academic content and complicates bibliography management. Therefore, understanding current researchers' practices and the challenges they face is necessary.

Previous studies have investigated current literature review practices, challenges [29], and "how" the HCI community cites papers [7, 35, 38]. Meanwhile, numerous tools and platforms have been proposed to assist researchers at different stages, including

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seeking information [5, 17], skimming papers [37], assisting critical reading [52], facilitating comprehension [26, 41], idea generation [19], and paper writing [9]. However, it remains unclear how researchers *forage*, *manage*, and *synthesize* citations, especially during prolonged overlaps between these phases. Gaining deeper insights into the challenges faced and the workflows adopted could guide the design of future systems.

To uncover researchers' current practices, challenges, and design implications when they *forage*, *manage*, and *synthesize* citations, we conducted semi-structured interviews with 12 researchers from the AI and HCI fields. Our thematic analysis revealed that (1) researchers are unable to fully track and digest all relevant work; (2) current citation management tools fall short in meeting researchers' needs; (3) "cherry-picking" is a common practice in citation synthesis; and (4) citation foraging, management, and synthesis workflows are disconnected and lack consistency. Our results also provide insights for the development of future interactive systems that will more effectively support researchers in their citation activities.

2 Related Work

2.1 Citation Foraging

Researchers typically conduct literature reviews and explore related work in two ways: top-down and bottom-up [25].

In the top-down workflow, researchers begin by obtaining an overview of the relevant literature based on search queries, then delve deeper into specific works. Digital libraries (e.g., ACM DL, IEEE Xplore) and research databases (e.g., Google Scholar [21] and Semantic Scholar [44]) enable access to published articles and support keyword-based queries to facilitate specific searches. Additionally, tools such as Connected Papers [47] and Citation Tree [10] assist the exploration of citations through interactive graphs.

In the bottom-up workflow, researchers focus on exploring related work starting from inline citations and citation chains. A typical approach consists of backtracking citations, but various systems have also been developed to enhance synthesis and comprehension when using this approach. For example, some systems augment inline citations [5, 53], connect follow-up papers with their predecessors [41], or help organize and explore inline citations based on user-defined threads [24]. For its part, Relatedly [36] is a literature review system that aids exploration by focusing on related work to help researchers discover and synthesize relevant studies. Other tools, like Synergi [25], aim to combine elements of both workflows to improve efficiency and provide a more holistic literature review experience.

However, while systems have been proposed to support both the top-down and bottom-up approaches, the subtleties behind citation foraging and how this task interacts with management and synthesis remain unclear.

2.2 Citation Management

Citation management sits at the intersection of foraging and synthesis workflows. Some studies, as discussed in Section *Citation Foraging 2.1*, have explored integrating foraging results into the citation management process [5, 24, 25, 36].

Tools like Zotero [43] and Mendeley [15] offer features such as the ability to create (sub-)folders to organize citations, add tags, and

employ active reading techniques like highlighting and annotating to enhance the understanding of papers. They also support shared libraries to facilitate the management of citations by coworkers. But bibliography management is also achieved using open file formats, such as Bibliographies are also frequently managed independently in external files. Typically, the BibTeX file [40] format is widely used as the reference to manage citations in LaTeX documents. Some commercial tools such as JabRef [48], EndNote [11], Zotero [43], Mendeley [15] are developed to assist the researchers in managing and editing such files.

That being said, the challenges faced by authors when managing citations remain unclear. Similarly, more research is needed on how to effectively integrate citation management with foraging and synthesis. Identifying these gaps is one of the primary motivations for our study.

2.3 Citation Synthesis

Showing the "true scholarly impact" [2] of one's work is one of the most important reasons for using citations. Citations play a crucial role in supporting authors' statements, whether for introducing, comparing, using, or extending related work [3, 50]. However, citation synthesis is a tedious task. As a result, several system [11, 15, 43, 48] mentioned in the previous Section *Citation Management 2.2* have explored to assist researchers better insert the citations.

Despite these advancements, citation practices are not always consistently followed. For example, Borg et al. [3] pointed out that newly graduated students often struggle to properly use source materials during academic writing. Bavelas et al. [2] found that some citations in academic papers are irrelevant, tangential, or even incorrect. In many cases, citations form chains across a series of papers [6, 39], as the number of publications increases, it becomes increasingly difficult for researchers to track the information they need [29]. Chen et al. [7] highlight that HCI research predominantly cites HCI papers, suggesting a potential focus on specific venues and disciplinary boundaries in citation practices. Similar to these works, the other motivation of our study aims to uncover the issues researchers face during the synthesis of citations.

3 Interview Study

To explore current practices, challenges, and design considerations related to citation foraging, management, and synthesis, we conducted semi-structured interviews with researchers of diverse experience. This study was approved by the Institutional Review Board (IRB) of the first author's institution.

3.1 Participants

We recruited 12 researchers (1 assistant professor, 2 postdoctoral researchers, and 9 PhD students) for this study. Demographic information is provided in the Appendix 2. All participants were recruited via a snowball sampling approach within the author's social network. All of them conducted research in the fields of HCI or AI, as researchers in these domains face significant publication growth and experience higher challenges in tracking, managing, and synthesizing citations.

3.2 Procedure

At the beginning of the interview, we introduced the purpose of our study. After participants confirmed their understanding of the research purpose and their participation, we invited them to recall the latest paper they had published and briefly describe its contents as a warm-up. Following this warm-up question, we proceeded with the interview following the prepared interview script (see Appendix 1). Each session lasted between 60 and 90 minutes, with a 10-minute break if the session exceeded 60 minutes. The interviews were conducted online in English via the Zoom platform. Audio recordings were made and deleted immediately after text transcription to ensure participants' privacy. The transcripts were coded, and thematic analysis [4] was conducted by the first author of the present document. Each participant was compensated approximately \$10 per hour for their time and participation in the interview.

4 Findings

4.1 Citation Foraging

4.1.1 Foraging citations is overwhelming. Nine participants (P1, P3, P5, P6, P7, P8, P9, P11, P12) expressed difficulties in searching for and filtering citations during their literature review. They noted that it is challenging to find appropriate citations, especially in rapidly evolving fields like "Artificial Intelligence" (P6). One participant further complained: "This is very time-consuming to search and filter the related work, especially considering there's a lot of such kind of papers" (P6). Three participants (P9, P11, P12) expressed that search engines may not provide the desired results. One of them commented, "It just feels like the papers you find on Google Scholar are not exactly what you're really looking for" (P12). Additionally, P9 and P11 mentioned that it is time-consuming to find citations whose content has a similar semantic meaning to their own statements, with P11 stating: "The hardest part is that you need to first find the three papers that have the meaning, the sentence, like which the meaning I need to prove in my paper" (P11).

Sometimes, these difficulties to find relevant situation results from a "cherry picking" writing approach we discuss in the following section 4.3.1, in which participants write statements first and then look for a citation to back it up, without knowing whether such a citation exists or not. Typically, P9 mentioned instances where they modified their statements because they could not find suitable supporting citations online.

4.1.2 Selective relevance in citation foraging. Participants described selecting citations based on multiple criteria, such as the venue, publication year, citation count, and abstract. Four Participants (P6, P8, P11, P12) reported the criteria to vary depending on the context or situation. For example, some stated, "For general, broad statements, we can use random papers as citations" (P6) and "I just pick the one that kind of supports my statement, and it's fine" (P8). Others relied on the sequence of search engine results, with one participant mentioning "I skim the first two pages of Google Scholar results [and select citations from them]" (P12).

Six participants (P1, P2, P4, P5, P7, P8) also expressed concerns about missing or misusing citations. However, five of them consider it is not a critical issue because such mistakes are "rarely pointed

out by reviewers" (P1) or could be easily corrected by adding the necessary citations if "the reviewers point it out" (P5). One of them also remarked, "No, no one cares whether I'm using the best [most suitable] papers to support my statement" (P8).

P11 mentioned that from the reviewer's perspective, the citations are not important, and he will not check the statement correctness of the citations while reviewing: "let me put some random or irrelevant citations in this paper. In this sentence, maybe I'm not very sure whether it happens. It is one of my concerns" (P11).

4.1.3 Lack of follow-up on research alert subscriptions. Three participants (P1, P3, P7) mentioned they subscribed to the research alerts to track the latest released paper, two of them subscribed to Google Alerts [20], and one subscribed to ResearchGate [42].

The motivation of such tools is to help researchers track the research trend, but interestingly, all of them reported they did not frequently check the research alert push notifications: "Sometimes it [Google Alerts] gives [me] too many [recommended] papers. And the alerts are too frequent. I cannot check them all... I just want them to give me the best. But when I checked the first [research alert] email, [they might] give me 10 results. [I just have patience] to see the title of the first three, then I do not want to read [the rest]" (P3).

P3 and P7 further described their wish to have more intelligent research alerts to access their citation management space, analyze their saved paper, reading history, and annotations to generate more precise recommendations: "[I wish to] add Google Scholar alert functions to the citation management. When I'm searching back [in my citation management space], I can get a notification on the trend on the searched direction and what the new findings are related. So it will be easier for me to connect to keep eyes on the new things" (P7).

4.2 Citation Management

4.2.1 Difficulties to remember citations, despite considerable efforts. During the literature review process, researchers invest time and effort in organizing citations in their personal management spaces to enhance sensemaking. However, dynamically managing large volumes of publications in such spaces remains a challenge. As a result, seven participants (P1, P5, P6, P7, P8, P9, P12) reported taking specific actions such as "Annotate", "Highlight", "Download", "Categorize" to support their memory and facilitate retrieval. Yet, participants reported to still forget the paper content after some time: "Though I tried to create categories to manage the papers. But it's too much. And I will forget everything [after some periods]" (P7).

Three participants (P5, P7, P12) mentioned that they saved papers locally or bookmarked them when they believed the articles could be relevant to their current research topics or useful for "future citation" (P8). Although they relied on citation management tools as external memory aids, they sometimes lost track of saved papers due to the "new papers released every day" (P7) or simply "forgot the existence of the paper" (P12). In other cases, participants recall reading and saving a paper and struggling to retrieve it later because they forgot its title or location: "I don't remember them at the final stage [when I need to insert citations in my draft]. So I need to re-find it: 'Oh, what is the name of that paper?'" (P5).

Besides saving paper in their citation management space, six participants (P1, P3, P6, P7, P8, P10) mentioned they are using or used highlight during the paper reading to enhance their comprehension.

However, they did not agree with the prolonged help of the highlight in the paper because they might forget the meaning behind the highlight: *“But sometimes you will forget where the highlight parts come from”* (P6).

Regarding how to resolve this issue, three participants (P1, P7, P10) expressed a desire for a reminder feature to help them recall papers they had read but may have forgotten. One participant suggested implementing a feature similar to the *“iOS previous photos remind feature”* (P1) to remind researchers of previously saved papers that they might have forgotten. Another participant mentioned, *“I wish citation management tools could have Google Scholar Alerts functions”* (P7), emphasizing the need for a system that integrates previous knowledge with newly released papers to aid in recalling prior content.

4.2.2 Need flexibility in citation management. *“Paper management for me is a hard task”* said P5. Eight participants (P3, P4, P5, P6, P7, P9, P10, P12) expressed dissatisfaction with the citation management tools they are currently using. Some participants complained that current commercial tools make it difficult to reorganize citations as they continually add papers to their collections: *“It’s difficult to think about the level of the tag. For example, if you put it in a tag called cv, and the amount of cv is very large, this tag cv becomes meaningless”* (P9).

This issue becomes more severe when researchers use a *“multi-tag system”* (P5) to manage their citations. On the other hand, participants also expressed the need to customize the sequence order and categories of papers in their management spaces: *“It’s kind of like sorting. So I can sort by name, the frequency I opened the document and the file sizes”* (P6).

4.3 Citation Synthesis

4.3.1 “Cherry-picking” citations to support claimed statement. All participants but P10 acknowledged that they habitually write citation statements even when they are unsure of the specific sources. They write the statement first, then search for supporting citations to strengthen its credibility and accuracy, eg. *“I need to prove the statement in the paper, so I need to find references to support the statement”* (P11). Regarding this cherry-picking behavior (write statement first, then look for supporting citations), four participants (P2, P5, P8, P11) expressed a desire to search for citations later, particularly for narrative or explanatory statements that help clarify a concept without affecting the main structure of the article: *“When I write some [statements] in my paper and I feel, oh, I need to cite something [to support that]. Then, like in the overleaf, I decided to do [literature review and] cite some paper for that”* (P2).

That being said, two participants (P8, P9) reported to search citation after writing statements but only when working on a discussion section, because the statements are generally the result of unexpected experiment results: *“Most of them [in the Discussion Section] are unexpected results, which means that you probably have no much knowledge on it, and you need to search more”* (P8).

4.3.2 Own publications reuse support. Two participants (P2, P11) who have already published several papers mentioned the habit of reusing the inline citations and citation statements from their previous publications. Therefore, they expressed the desire to reorganize

previously created bibliography files directly in the software they use for writing the new paper: *“I normally like to put a lot of citations in that [bib file] and also download the previous project citation file and put in my new project. So, I know that my Overleaf project has a big pool. If Overleaf can provide that kind of thing (directly managing citations in the overleaf), I think I’m happy to use that”* (P2).

4.4 Interactions Between Phases

4.4.1 Fragmented tool synthesis in literature activities. Literature-related activities are sophisticated and consist of several key steps, such as information retrieval, paper reading, sensemaking, and paper synthesis. Various platforms and software tools support different parts of these processes. In our user study, seven participants (P2, P5, P7, P8, P9, P10, P11) expressed a desire for smoother integration and collaboration between each of these literature activities: *“But the issue is that I need to go to my Google Drive, go to my folder, find the conference sub-folder, and then open that PDF, and then drag the title, copy and paste into Google Scholar, and then export the .bib text file, and then copy paste on my like Overleaf”* (P8). In another participant’s synthesis case, they managed the papers’ titles in one software and the notes in other documents. In this situation, they need to frequently cross-refer different tools to check the paper’s info. P7 pointed out that they wish to *“link the personal notebook and the original location in the paper”*.

Two participants (P2, P11) complained about the synthesis of bibtex files because they faced difficulties locating the correct paper they wanted to cite: *“So you can find that there’s actually no keywords or know what it’s actually doing. There’s just a title. So as I mentioned, how could I use this?”* (P11).

4.4.2 Optimistic yet cautious attitude toward collaborating with AI. Most participants reported using generative AI tools including ChatGPT [34], STORM [23, 45], PDF Expert [49], in their research activities, such as for paper memorization and recommendations, and expressed a positive attitude toward these tools. One participant noted utilizing ChatGPT to generate search queries: *“It’s already very helpful. If at least one out of the five keywords is usable, that’s already enough.”* (P8). One expressed a desire for a tool that could recommend supportive citations from papers they had read but could not recall precisely: *“Let’s say when I’m writing, I want to quickly check something I’ve read before.”* (P7). Another suggestion was to create a related work writing guideline system to help researchers synthesize their saved related works: *“You can give me a guideline for how to write a paper. For example, I already have some related papers. What I want is a tool that can help me organize those papers.”* (P5).

However, eight participants (P1, P3, P5, P6, P8, P10, P11, P12) expressed that they did not fully trust the final results generated by AI. Instead, they preferred to see specific *“semantic matching sentences”* (P11) within the paper to verify the generated results. For example, P11 further explained: *“[I] have a wish that if I want to prove a sentence, I just ask the ChatGPT, like, ‘could you find some [sentences from the] citation that can prove the following sentence?’, and I will select the suitable sentence I want [to support my sentence]”* (P11).

5 Design Implications

5.1 Intelligent Research Alerts

Research alert tools loosely connect researchers with their research contexts, as these systems [21] typically proactively push the latest paper notifications to subscribers' emails. Previous work has improved this problem. For instance, PaperWeaver [27] is a system that can provide paper recommendation alerts based on the user's own research libraries, we propose that the future alert system can intelligently push notification alerts based on the user's different usage scenarios and different usage habits. For example, the system could push different kinds of alerts based on a researcher's daily schedule, such as pushing new related papers to the researcher on the way to the office in the morning, but push the content have already read to help review the key points when back home. To implement this function, we can adopt the design methods of Just-in-Time Adaptive Interventions (JITAI) [30] to deliver adaptive research alerts based on the researcher's environmental state.

In addition, we need solutions to assist researchers in following content alerts. Traditional PDF-format papers are difficult to read on small devices while on the move. Recently, NotebookLM [22] proposed converting papers into audio summaries, enabling users to listen to research content while on the go. However, users cannot interact with audio content and may easily lose focus. They also lose access to most visual illustrations that can be provided. While active reading techniques, such as highlighting and annotations, enhance the reading experience, we should further explore "active listening" methods to improve audio-based learning experiences.

5.2 Dynamic Citation Management

With the increasing number of publications, researchers face challenges in managing their citations. In the study, we found that researchers wish to have intelligent citation management tools, such as flexibly reorganizing of citation categories. Previous work has explored solutions to classify related work during the academic reading process, the system can help researchers integrate the inline citations and related work in their own management space [24]. Based on that, we suggest that future citation management tools could semi-automatically categorize the literature during the literature review and update when the researcher's interest changes. For example, if the user meets difficulty to classify some paper in the existent categories, they can input some keywords to let the system re-organize the libraries based on that. Additionally, the management space could be "more intelligent" to integrate existing papers saved in the management system with newly published papers that the user has not read yet.

5.3 Recall Forgotten Memories

Conducting literature reviews and related tasks is often tedious and challenging. From our formative study, we learned that some participants encountered difficulties in recalling papers they had previously read. Although they remembered its content, they were unable to locate it. It took significant time and effort for them to retrieve such papers. Future tools should help researchers to track their reading histories and manage their sensemaking space, making it easier for them to retrieve citations.

Current citation management tools also normally provide active reading features such as highlighting and annotations [15, 43]. These features accumulate rich data to analyze users' research behaviors. Future citation management tools should actively utilize these data. For example, the system can integrate highlights and notes into the current knowledge context, helping users better recall their previous comprehension of specific papers.

5.4 Critical Citing, Not Cherry Picking

With the growing number of publications, various generative AI tools have emerged to assist researchers in conducting literature reviews [33] and skimming papers [17]. However, these tools may contribute to "cherry-picking" citations and enhance the citation injustice [12, 13] to the research community. Moreover, they may mitigate the researchers' critical thinking ability [1]. In the future, designers should carefully develop research support tools to prevent users from over-relying on AI and to encourage researchers to critically read papers and cite references.

5.5 Good Citation Practice Sharing in Research Communities

We found that researchers often adopt selective citation practices without careful comprehension, which may be attributed to the rapid increase in publications and the growing number of citations in papers. Additionally, reviewers face immense pressure to verify all citations. Citations should serve as a means to uphold the integrity of scientific communication and a bridge for readers and authors collaboratively verify the presented evidence [32], however, we identified a lack of transparency between authors and reviewers. This issue is particularly evident in closed peer reviews, where reviewers cannot thoroughly verify all citations in a draft to ensure consistency between the author's statements and the cited papers.

ReviewFlow [46] has explored showing a general abstract of inline citations and related work to help reviewers check the inline citations, we further suggest authors could be required to explain why they selected specific citations to support their claims and to indicate where the supporting evidence (e.g., specific sections or sentences) is located. This would provide better support for the review process and enhance the overall quality of citation practices.

6 Limitation and Future Work

While our work explores the practices, challenges, and design implications for citation foraging, management, and synthesis, in the early phase of the research, there are some limitations in this study that future work should address.

First, we conducted a small-group, semi-structured interview with 12 young researchers as participants from constrained fields—HCI and AI. We propose to expand both the sample size and diversity of participants to compare potential differences in citation practices across HCI, AI, and other fields to point out what the HCI and AI community can learn from other disciplines. The thematic analysis should be conducted collaboratively by multiple authors and reviewed by external experts to ensure rigor.

Then, the discussion of the current study focused more on the citation practice challenges and design implications in each citation phase. The transition-specific challenges and design implications

across citation foraging, management, and synthesis require further exploration and discussion in future work.

Finally, in future work, we will further engage in a thorough discussion to problematize the implications of different citation practices.

7 Conclusion

With the surge of publications, researchers face challenges in citation foraging, management and synthesis. To understand researchers' current practices and challenges, we conducted semi-structured interviews with 12 experienced researchers from the AI and HCI fields. Our key findings include: (1) researchers are unable to fully track and digest all relevant work; (2) current citation management tools fall short in meeting researchers' needs; (3) "cherry-picking" is a common practice in citation synthesis; and (4) citation foraging, management and synthesis workflows are disconnected and lack consistency. Our design implications provide insights for the development of interactive systems that more effectively support researchers in their citation activities.

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A Interview Script

Table 1: Interview Script

No.	Question
1	How many years of experience do you have in your research field?
2	Please briefly introduce your recently published academic paper.
3	You have XXX references in total in your recent paper. How do you feel about this number?
4	Was it the right amount, too much, or too little? Why do you think so?
5	If you feel it is too much, what creates such a feeling? How do you think you can reduce your workload?
6	You have read YYY papers for this project. How did you select XXX papers to use as references from the YYY papers?
7	How long do you think you spend filtering these XXX references?
8	How do you think you can reduce your workload?
9	How did you manage the XXX references? Why did you choose that method?
10	What is your goal in managing the references?
11	What information did you collect from each paper? Why did you collect it?
12	Do you use any tools for reference management? What do you think of these tools? Why do you think they are good or not good to use?
13	What features would you like to add to these tools?
14	Where is the most overwhelming or tedious part of your literature-related activities?
15	Do you think you will need to cite more references in your future papers? Why do you think so?
16	How do you think you would approach similar tasks in your future papers?
17	Do you foresee any challenges with your current approach? If so, what would be the most difficult or painful part to overcome?
18	Is there anything you would like to share with us?

B Study Participants

Table 2: Participant Information

ID	Age	Gender	Years of Experience	Current Role	Major
P1	25	M	6	PhD Student	HCI
P2	29	M	5	Assistant Professor	HCI, ML
P3	30	M	7	PhD Student	HCI, NLP, CV
P4	25	M	4	PhD Student	CV
P5	27	M	4	PhD Student	HCI, Visualization
P6	27	M	5	PhD Student	CV, CG
P7	26	F	1.5	PhD Student	HCI
P8	28	M	6	Post Doctor	HCI, NLP, CV, ML
P9	26	M	6	PhD Student	HCI
P10	31	F	9	Post Doctor	HCI, CV, CG
P11	27	M	5	PhD Student	HCI
P12	27	F	5	PhD Student	HCI